APPLICATION · FILM AND FOIL

EDGE POSITION REFERENCE IN FILM AND FOIL LINES

In film and foil lines, everything starts with where the two edges will sit. Reel width and cut width change from recipe to recipe, so the edge position has to be re-established at every changeover. ServoMotion moves the two edge lines to the targets read from the recipe, and the positions stay visible while the line runs.



Core	K3 single channel + fixed	2,000 mm
SYSTEM USED IN THE EXAMPLE INSTALLATION	1 MOVING + 2 FIXED REFERENCES	TRAVEL

ON THE LINE — DO YOU RECOGNISE THESE SYMPTOMS?

A new edge on every reel change

When the reel width changes, the edge position is measured from scratch, making setup the longest item in the changeover.

Transparent material fools the eye

On a thin, glossy surface, where the edge actually ends depends on the viewing angle. The same recipe is set up differently across shifts, resulting in inconsistent output.

Constant edge scrap

Running with a safety margin adds a little scrap to every metre, and that adds up to a meaningful figure by the end of the year.

Mechanical guides drift

Fixed guides shift position with vibration and need periodic readjustment.

WHAT WE MEASURE DURING THE SITE SURVEY

Reel width range	The difference between the narrowest and widest recipe determines travel range and housing length.
Edge tolerance	The accepted deviation; line thickness and working distance are selected accordingly.
Material surface	Transparent, matte or metallised surfaces affect laser colour and optical power.
Working distance	The vertical distance between the laser and the material determines angular tolerance.
Line speed	Determines how quickly a position change has to complete.
Support structure	An existing traverse or a new frame.

WHICH ARCHITECTURE FITS THIS SEGMENT?

MOST APPLICATIONS

CORE PRECISOR

In film and foil lines, the number of simultaneous reference lines is usually limited to the two edges. An enclosed industrial profile drives to the target positions under servo control, while fixed references keep the control points beyond the edges continuously visible.

IF THE NUMBER OF PROJECTED LASER LINES GROWS

SHUTTLE PRECISOR

If many slit references have to be visible at once, the shuttle architecture takes over. The decision is based on the densest recipe in your portfolio.

EXAMPLE INSTALLATION

CORE PRECISOR · K3 · 2,000 MM TRAVEL

One moving channel and two fixed references. The moving channel travels to the edge position defined in the recipe, while the fixed lines keep the control points visible throughout production. These figures come from one example installation; the dimensions for your line are determined during the site survey.

This note summarises the typical requirements of the segment and one example installation. The final system architecture, housing size and scope of connection are established during the site survey. The lasers provide an alignment reference; material and machine elements are set by the operator. Values may change without notice as the products are developed.

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